Diffusion Feature Completion-Driven Textual Dual Distillation for Multimodal Remote Sensing Classification With Missing Modality

Dan Xu, Wenqian Dong, *Senior Member, IEEE*, Song Xiao, Jiahui Qu, *Senior Member, IEEE*, and Yunsong Li, *Member, IEEE*

***Abstract*—In recent years, research on joint classification of multimodal remote sensing data has achieved remarkable progress. However, constrained by imaging conditions or sensor issues, missing modality frequently occurs in practice. Knowledge distillation, as one of the mainstream strategies to deal with missing modality, is widely used due to its excellent transfer ability. However, most existing knowledge distillation methods focus on directly transferring available modal information through feature alignment or predictive constraints, without explicitly modeling or refining the latent semantic structure of missing modality. Furthermore, the heterogeneity between different modalities makes effective alignment difficult through direct transfer, thus limiting the utilization of cross-modal complementary information and consequently hindering model performance improvement in remote sensing scenarios with missing modality. Therefore, in this article, we propose a diffusion feature completion-driven textual dual distillation model (DFC-TD^2) for multimodal remote sensing image classification with missing modality. This model explicitly models the missing modality, introduces textual semantic information to guide cross-modal feature transfer, and achieves multiconstraint collaborative optimization under the supervision of classification labels, thereby effectively improving the overall performance of multimodal classification tasks with missing modality. Specifically, a shared-constrained diffusion feature completion network (SCFC) is designed, which uses shared features extracted and frozen from multimodal data as constraints to guide the diffusion model to effectively model and complete missing modal features, thereby generating semantically consistent and more discriminative modal representations. Furthermore, a text-guided dual distillation network (TGD^2) is designed to guide cross-modal feature alignment using text semantics and combine it with classification label supervision to construct a feature-label dual constraint, which effectively improves feature discriminativeness and supervision efficiency, thereby mitigating cross-modal representation heterogeneity and improving the stability and effectiveness of knowledge transfer. Experimental results on three public datasets demonstrate the effectiveness of this method in dealing with missing modality. The code is available at: https://github.com/Jiahuiqu/DFC-TD2.**

***Index Terms*—Image classification, knowledge distillation, latent diffusion models (LDMs), missing modality, multimodal, remote sensing.**

I. Introduction

MULTIMODAL remote sensing imagery integrates heterogeneous data characterized by different spatial resolutions, spectral bands, and imaging mechanisms, enabling complementary utilization of multimodal information, thus providing a more comprehensive and 3-D representation of ground targets, and plays an irreplaceable role in tasks such as land cover analysis and utilization [1], [2], object detection [3], [4], change detection [5], [6], [7]. In multimodal remote sensing classification, existing studies mainly focus on effective feature extraction and fusion to exploit complementary information across modalities for improved accuracy and robustness. Early methods based on convolutional neural networks (CNNs) capture spectral and spatial features through local receptive fields and weight sharing [8], [9], [10]. Transformer-based models further enhance representation by leveraging self-attention to model long-range dependencies and global context, improving cross-modal alignment [11], [12]. More recently, Mamba architectures built on state space models (SSMs) have demonstrated strong capability in jointly modeling multiscale spatial features, spectral information, and cross-modal interactions, effectively addressing modal heterogeneity [13], [14]. In addition, textual information has been introduced as semantic guidance to facilitate cross-modal understanding and further improve classification performance [15]. With the development of research, multisource feature fusion strategies have been used for joint modeling of cross-modal data to improve representation capabilities. For example, attention mechanisms further enhance discriminative representations by capturing multidirectional and multiscale feature interactions in complex scenes [16], [17]. Moreover, Wang et al. [18], [19] improved feature consistency and discriminative power via graph regularization and distribution constraints, effectively mitigating cross-modal distribution discrepancies and offering a new perspective for this field.

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Dan Xu, Wenqian Dong, Jiahui Qu, and Yunsong Li are with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: xdan@stu.xidian.edu.cn; wqdong@xidian.edu.cn; jhqu@xidian.edu.cn; ysli@mail.xidian.edu.cn).

Song Xiao is with the Department of Electronic and Communication Engineering, Beijing Electronic Science and Technology Institute, Beijing 100070, China, and also with the School of Telecommunications Engineering, Xidian University, Xi'an 710071, China (e-mail: xs_xidian@163.com).

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Current mainstream multimodal remote sensing classification methods typically assume that the modal information input during the training and testing phases remains consistent and rely on complete modal data for joint modeling. However, in practical applications, obtaining complete and high-quality multimodal remote sensing data is often challenging due to factors such as sensor failures, inconsistent satellite revisit cycles, and data loss during acquisition or transmission, resulting in widespread modality missing data [20]. When a modality becomes unavailable, the existing cross-modal associations and complementarities are disrupted, leading to a significant decrease in classification performance. Therefore, how to achieve efficient and robust multimodal joint classification under missing modality conditions has become one of the key problems that urgently need to be solved in the field of multimodal remote sensing image understanding.

Mainstream methods for handling the joint multimodal classification problem with missing modality can be divided into two main categories: 1) based on generative models and 2) based on knowledge distillation [21], [22]. In the category based on generative models, one approach is to learn shared features across modalities to generate missing data [23], [24]. These methods typically utilize available modality to extract high-level semantic representations or latent features, and map different modalities to a unified shared subspace. This allows for the reconstruction or approximation of data or feature representations of a missing modality based on the shared features. However, this modeling approach, centered on shared representations, to some extent ignores the discriminative information unique to each modality, weakening the complementary features between modalities and consequently negatively impacting classification performance. Another approach is to directly generate the missing data using generative models [25]. These methods typically use existing modality as conditions and directly fill in the missing modal features through a generative model. However, the instability of the generation process can easily introduce semantic bias and noise, thus affecting classification performance.

In the category based on knowledge distillation, the core idea is to transfer discriminative knowledge from a well-trained teacher model with access to all modalities to a student model operating under missing modality conditions, thereby maintaining robust performance even when a particular modality is unavailable [26]. Existing knowledge distillation-based methods for dealing with the missing modality can be mainly divided into the following categories, as shown in Fig. 1. However, the significant heterogeneity between different modalities poses a serious challenge to cross-modal knowledge alignment, which can easily lead to the loss of key information. At the same time, traditional knowledge distillation methods usually do not explicitly model missing modality, making it difficult to preserve the modality-specific information representation.

To overcome the inherent limitations of knowledge distillation methods in modality-missing scenarios, we propose a diffusion feature completion-driven textual dual distillation model (DFC-TD^2) for multimodal remote sensing image classification. This model enhances multimodal representation

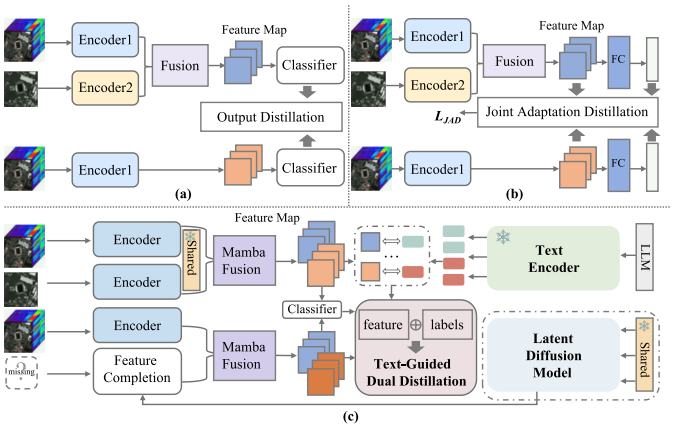


Fig. 1. Main methods of knowledge distillation to deal with missing modality. (a) Single distillation. (b) DD. (c) Proposed method: a text-guided DD model driven by diffusion FC.

capabilities by reconstructing missing modality features and introduces a text-guided dual knowledge distillation mechanism to promote cross-modal knowledge alignment and transfer, thereby effectively mitigating the impact of missing modality on multimodal joint classification performance. Specifically, to generate high-quality and semantically consistent missing modal features, we designed a shared-constrained diffusion feature completion network (SCFC). This network introduces shared features extracted and frozen from multimodal data as semantic constraints to guide the diffusion process in the latent space to model and reconstruct missing modality, thereby generating semantically consistent feature representations with stronger discriminative power. Simultaneously, this network provides more reliable feature input for subsequent text-guided dual distillation (DD), alleviating the limitations of traditional distillation methods and further improving overall performance. Furthermore, we designed a text-guided DD network (TGD^2), which introduces textual information as semantic guidance to achieve cross-modal knowledge transfer, thereby effectively mitigating the heterogeneity differences between modalities. This network deeply couples feature layer distillation with classification label supervision, constructing a multilevel DD mechanism. This significantly enhances feature discriminative power while improving the efficiency of supervision signal utilization, thus effectively supporting high-precision multimodal classification with missing modality. Unlike traditional knowledge distillation methods that ignore missing modality, this article proposes a network that explicitly models and completes the feature information of missing modality, and introduces textual semantics as a guide for cross-modal knowledge transfer. This effectively promotes information interaction and knowledge alignment between modalities, significantly improving the performance of multimodal remote sensing classification with missing modality. To summarize, the contributions of this work are as follows.

1) We propose a DFC-TD^2 to achieve multimodal remote sensing joint classification with missing modality. This model explicitly models and completes the missing modal features, introduces textual semantic information

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as a high-level prior to guide cross-modal feature transfer, and forms a dual constraint mechanism together with classification label supervision, thereby significantly improving classification performance with missing modality.

2) We design an SCFC, which uses multimodal shared features as conditional constraints to generate and complete missing modality in the feature space in a constrained manner through a latent diffusion model (LDM), thereby effectively ensuring the semantic consistency and stability of the completed features.
3) We present a TGD^2, in which text semantics guides cross-modal feature distillation and classification labels supervise output-level distillation, thus achieving effective multimodal knowledge transfer and representation learning in the missing modality.

II. Related Work

A. Multimodal Remote Sensing Classification

Multimodal remote sensing classification methods aim to fully utilize and effectively integrate complementary information between different modalities, thereby achieving high-precision identification and classification of ground cover categories. Current research predominantly employs CNNs, transformers, and their hybrid architectures to enhance multimodal feature modeling and fusion. These approaches exhibit strong representation capabilities and have gradually become the mainstream paradigms in multimodal remote sensing image classification [27], [28], [29].

As model size and sequence length continue to grow, SSMs have attracted widespread attention due to their efficiency and stability in complex sequence modeling, with the Mamba architecture being a particularly noteworthy approach [30], [31]. MSFMamba [32] introduces a Mamba-based unified modeling framework that jointly implements multiscale spatial modeling, spectral feature extraction, and cross-modal feature fusion within the same system, effectively bridging the heterogeneous gap between modalities, improving computational efficiency while further enhancing classification performance. Existing models typically rely on complete modal inputs, but in practical scenarios, missing modalities are common, which can significantly degrade performance. Therefore, developing multimodal classification models that are robust to missing modality is of critical importance.

B. Generative Model

For multimodal joint classification with missing modality, a representative approach involves reconstructing the missing modality using generative models. These methods typically employ available modality as conditional inputs to synthesize data or feature representations of the missing modality. Generative adversarial networks (GAN) are widely adopted due to their strong generation capabilities [33]. For example, CM-GAN [34] reconstructs missing modality from existing ones, effectively recovering long-term time series even when an entire modality is absent. Similarly, the multiple-input multiple-output GAN framework [35] leverages redundancy and complementarity among multimodal sequences to generate one or more missing modalities simultaneously, addressing diverse missing modality scenarios. However, the inherent instability of adversarial training makes GANs prone to noise and mode collapse, limiting their effectiveness in high-precision classification tasks.

With the advancement of generative modeling, diffusion models have been increasingly introduced into missing-modality reconstruction due to their stable training and superior generation quality. These models iteratively transform Gaussian noise into target data through forward diffusion and reverse denoising processes. The denoising diffusion probabilistic model (DDPM) demonstrates strong performance in semantic feature modeling, denoising, and image restoration [36]. Furthermore, the LDM conducts diffusion in a low-dimensional latent space, significantly reducing computational cost while improving generation efficiency, and has been widely applied in high-resolution synthesis, denoising, and restoration tasks [37], [38]. Consequently, LDM provides robust support for high-quality and efficient feature completion (FC) in missing-modality scenarios.

C. Knowledge Distillation

For multimodal joint classification problems under missing modality conditions, a representative approach is based on knowledge distillation for cross-modal knowledge transfer. Knowledge distillation transfers knowledge from a teacher to a student model, maintaining high performance with reduced complexity, and is widely used to address missing modality in cross-modal tasks. A more robust classification model is obtained by performing a single online distillation on the output labels [39]. The DGDNet framework [40] utilizes a teacher–student network structure to transfer multimodal information to the student network, and mitigates representational heterogeneity and insufficient training of weak modality through diversity-guided distillation and modality-centric regularization, achieving high-performance inference even with multiple missing modalities. Building upon this, DH-ADNet [41] combines multilayer attention distillation with multimodal collaborative sample selection to learn discriminative features and privileged information during the training phase, enabling the model to maintain accurate land cover classification performance even when a modality is missing. The prompt framework uses pretrained text features to guide unsupervised knowledge transfer [42]. Inspired by the above methods, this article introduces text semantics-guided feature transfer and combines it with output-level supervision to improve feature alignment and knowledge transfer capabilities.

III. Methods

A. Overview

In order to maintain simplicity and universality, a two-modality image dataset is defined, expressed as $M = \{\mathbf{X}_{m1}, \mathbf{X}_{m2}, y\}$, where $\mathbf{X}_{m1}$ and $\mathbf{X}_{m2}$ correspond to the images of modalities m_1 and m_2, respectively, while y represents the category label of the images. In practical applications, data

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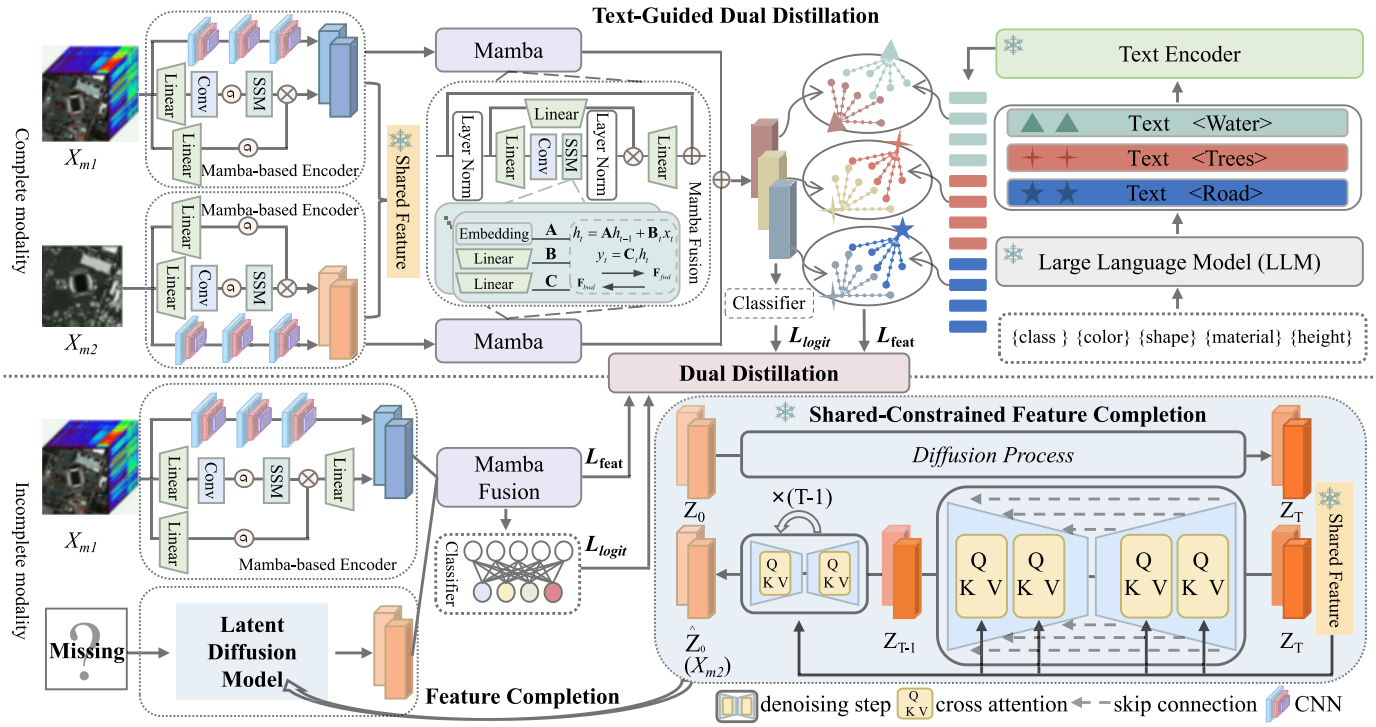


Fig. 2. Overall architecture of the proposed DFC-TD^2. This method consists of two networks: 1) a SCFC that conditions on multimodal shared features to complete the features of the missing modality via LDM, and 2) a TGD^2 that leverages textual semantics for feature distillation and category-level distillation.

from certain modalities are often missing due to sensor malfunctions, inconsistent acquisition times, or data corruption. Therefore, datasets may not always contain data from all modalities. A dataset containing data from only one modality is represented as $M_1 = \{\mathbf{X}_{m1}, y\}$ or $M_2 = \{\mathbf{X}_{m2}, y\}$. This article aims to fully exploit and utilize the complete modal information in multimodal data during the training phase. By learning the complementary relationships and potential correlations between different modalities, the model can maintain accurate category prediction and robust performance during the testing phase, regardless of whether the modalities are complete or not. To achieve this goal, we propose the DFC-TD^2 model, driven by modal FC and integrating text-guided feature transfer with classification supervision. Specifically, it reconstructs missing modality to recover richer modality-specific information, while leveraging textual semantics to enable effective cross-modal alignment. The joint constraints of feature-level and label-level supervision optimize the knowledge transfer process and significantly improve the robustness and classification performance of the model in the missing modality.

The proposed DFC-TD^2 model architecture is shown in Fig. 2, where the complete modality serves as the main teacher network, the missing modality completion serves as the auxiliary teacher network, and the incomplete modality serves as the student network. To achieve high-quality completion of missing modal information, we design an SCFC. This network extracts and freezes multimodal shared features from the main teacher model, uses them as conditional constraints to guide the reconstruction of missing modal features, and performs feature-level completion within the framework of the LDM, thereby obtaining more accurate and stable modal feature representations and achieving effective recovery and enhancement of missing information. Subsequently, in the TGD^2, text semantics are introduced to enhance the feature transfer capability of the main teacher model. While performing feature-level distillation, output layer supervision is combined with classification labels provided by the teacher model to achieve class-level distillation. The completion of modal features provides rich and high-quality feature information for cross-modal knowledge transfer, while the text-guided multilevel DD mechanism effectively promotes knowledge transfer between the teacher and student networks, thereby significantly improving classification performance and model robustness in cases of missing modality.

B. Shared-Constrained Diffusion FC

To alleviate the limitations of traditional knowledge distillation methods in modeling modal information when modality is missing, and to further obtain richer and more discriminative modal feature representations, this article proposes an SCFC. This network uses multimodal shared features as constraints to guide the completion process of missing modality features, generating high-quality feature representations that are semantically consistent and more discriminative. By effectively modeling and completing missing modality, this method significantly alleviates the information loss caused by incomplete modality, providing more stable and reliable feature support for subsequent cross-modal knowledge transfer and classification tasks.

Specifically, multimodal shared features are learned by projecting features from different modalities into a unified

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low-dimensional semantic subspace, and a central moment difference constraint is introduced to align the features of each modality at the statistical distribution level. By simultaneously constraining the statistical consistency of different modalities at the mean, variance, and higher order central moment levels, the features projected into the shared subspace become highly consistent in their distribution structure, thus characterizing and forming a multimodal shared feature representation. Based on this, the obtained shared feature $\mathbf{C} = \mathbf{F}_{\text{shared}}$ is frozen and used as a conditional constraint, providing stable, consistent, and semantically reliable cross-modal guidance and constraints for missing modality during the FC process. Regarding the training strategy, the SCFC network is first pretrained on complete multimodal data to learn a unified shared latent representation and acquire reliable feature reconstruction capability. After this independent pretraining stage, the SCFC network is fixed and directly integrated into the subsequent text-guided DD framework for missing modality FC, thereby ensuring stable training without introducing additional optimization burden. During inference, when a certain modality is missing, the pretrained SCFC relies solely on the available modality and the learned prior knowledge from the diffusion model to guide latent diffusion-based FC.

Different from image-based LDMs, the diffusion process in this method takes place in the feature space. This design allows for focusing on high-level semantic representations of modal features while reducing computational complexity and sensitivity to low-level pixel noise, thereby improving the discriminative ability and stability of the completed features. Specifically, the forward process adds Gaussian noise to the input feature $\mathbf{Z}_0$ in a Markov chain manner, gradually transforming it into a noise sample $\mathbf{Z}_T$ close to a standard Gaussian distribution. The conditional transition at each time step t is defined as

$$q(\mathbf{Z}_t \mid \mathbf{Z}_{t-1}) = \mathcal{N}(\mathbf{Z}_t;\ \sqrt{1-\beta_t}\,\mathbf{Z}_{t-1}, \beta_t \mathbf{I}) \tag{1}$$

where $t \in [0, T]$, and $\mathcal{N}(\mathbf{Z}_t; (1-\beta_t)^{1/2}\,\mathbf{Z}_{t-1}, \beta_t \mathbf{I})$ denotes that the random variable $\mathbf{Z}_t$ follows a Gaussian distribution with mean $(1-\beta_t)^{1/2}\,\mathbf{Z}_{t-1}$ and covariance $\beta_t \mathbf{I}$. Here, $\beta_t \in (0, 1)$ is the noise intensity at step t. Based on this formulation, after multiple iterations, $\mathbf{Z}_t$ can be obtained directly as

$$\mathbf{Z}_t = \sqrt{\bar{\gamma}_t}\,\mathbf{Z}_0 + \sqrt{1-\bar{\gamma}_t}\,\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}) \tag{2}$$

where $\gamma_t = 1-\beta_t$ and $\bar{\gamma}_t = \prod_{i=0}^{t} \gamma_i$. Here, $\mathbf{Z}_T \sim \mathcal{N}(0, \mathbf{I})$ denotes the fully noised latent representation.

The reverse denoising process utilizes both the noisy latent state $\mathbf{Z}_t$ and shared features $\mathbf{C}$ as conditional inputs to reconstruct high-quality features $\hat{\mathbf{Z}}_0$. Within this process, the posterior distribution $p(\mathbf{Z}_{t-1} \mid \mathbf{Z}_t, \mathbf{Z}_0)$ plays a fundamental role

$$p(\mathbf{Z}_{t-1} \mid \mathbf{Z}_t, \mathbf{Z}_0) = \mathcal{N}(\mathbf{Z}_{t-1}; \mu(\mathbf{Z}_t, \mathbf{Z}_0), \Sigma(\mathbf{Z}_t, \mathbf{Z}_0)) \tag{3}$$

where $\mu(\mathbf{Z}_t, \mathbf{Z}_0)$ is the mean and $\Sigma(\mathbf{Z}_t, \mathbf{Z}_0)$ is the variance, typically represented as

$$\begin{aligned} \mu(\mathbf{Z}_t, \mathbf{Z}_0) &= \frac{\sqrt{\gamma_t}(1-\bar{\gamma}_{t-1})}{1-\bar{\gamma}_t}\,\mathbf{Z}_t + \frac{\sqrt{\gamma_{t-1}}(1-\gamma_t)}{1-\bar{\gamma}_t}\,\mathbf{Z}_0 \\ \Sigma(\mathbf{Z}_t, \mathbf{Z}_0) &= \frac{(1-\bar{\gamma}_{t-1})(1-\gamma_t)}{1-\bar{\gamma}_t}. \end{aligned} \tag{4}$$

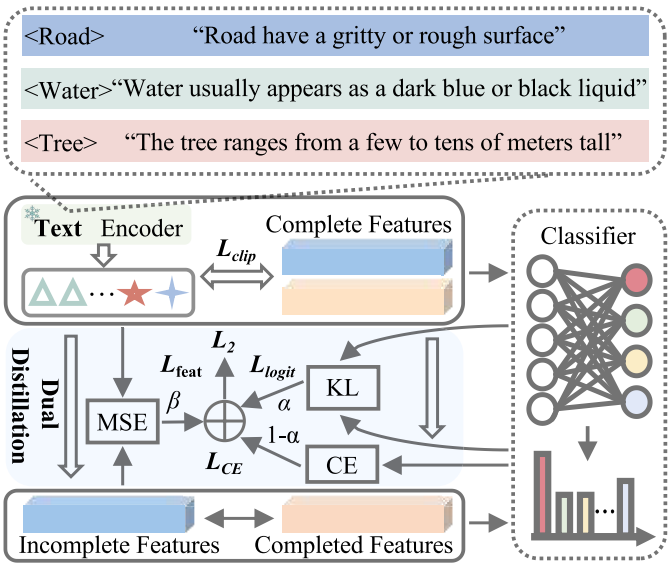


Fig. 3. Overall architecture of the proposed TGD^2.

The posterior mean and variance guide the sampling of the latent state at time step $t-1$, yielding the denoised feature as

$$\begin{aligned} \hat{\mathbf{Z}}_{t-1} = &\sqrt{\frac{\bar{\gamma}_{t-1}(1-\gamma_t)}{1-\bar{\gamma}_t}}\,f_\theta(\mathbf{Z}_t, \mathbf{C}) + \sqrt{\frac{\gamma_t(1-\bar{\gamma}_{t-1})}{1-\bar{\gamma}_t}}\,\mathbf{Z}_t \\ &+ \sqrt{\frac{(1-\bar{\gamma}_{t-1})(1-\gamma_t)}{1-\bar{\gamma}_t}}\,\boldsymbol{\epsilon}. \end{aligned} \tag{5}$$

Here, f_θ denotes the denoising neural network parameterized by θ, which takes as input the noisy feature $\mathbf{Z}_t$ at time step t. The network progressively denoises features from the initial noisy state at $t = T$ to $t = 1$, guided by predicted clean features, ultimately reconstructing $\hat{\mathbf{Z}}_0$. In this way, the reverse denoising process is completed, reconstructing the missing modal features while enhancing the feature representation capabilities, thus providing high-quality modal feature support for the subsequent distillation process.

C. Text-Guided DD

To enhance cross-modal knowledge transfer capabilities, this article proposes a TGD^2, as shown in Fig. 3. At the feature level, the network uses textual semantic information as guidance to explicitly reduce the heterogeneity of features from different modalities in the representation space, achieving refinement and alignment of cross-modal features. At the output layer, it uses classification labels provided by the teacher model for supervision, achieving category-level distillation and further strengthening the discriminative ability of the student network. The entire process is optimized end-to-end through a unified loss function, enabling feature-level and category-level distillation to work collaboratively, effectively improving knowledge transfer efficiency and classification performance in cases of missing modality.

The Mamba architecture, based on a selective state-space modeling mechanism, provides scalability and efficiency for remote sensing image processing. Its ability to model long-range dependencies and adapt to heterogeneous patterns makes

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it particularly suitable for multimodal remote sensing classification. As shown in Fig. 2, the proposed full-modal teacher network uses a Mamba-based encoder for each modality, followed by CNN-based refinement to enhance the local spatial representation. From the perspective of feature modeling, the core state update of the SSM is formulated as

$$\begin{aligned} h_t &= \mathbf{A}h_{t-1} + \mathbf{B}_t x_t \\ y_t &= \mathbf{C}_t h_t \end{aligned} \tag{6}$$

where h_t denotes the hidden state at time step t, x_t represents the projected input feature, $\mathbf{A}$ is the state transition matrix, and $\mathbf{B}_t$ and $\mathbf{C}_t$ are input-dependent SSM parameters dynamically generated at each time step.

To further enhance sequential context modeling, the SSM performs state propagation in both forward and backward directions, and the resulting features are fused as

$$\mathbf{F}_{\text{SSM}} = \mathbf{F}_{\text{fwd}} + \text{Reverse}\left(\mathbf{F}_{\text{bwd}}\right). \tag{7}$$

Here, $\mathbf{F}_{\text{fwd}}$ and $\mathbf{F}_{\text{bwd}}$ denote the forward and backward SSM outputs, respectively, while Reverse$(\cdot)$ represents the sequence reversal operation. Through bidirectional modeling, the network captures contextual dependencies in both directions, enabling a more comprehensive representation of global semantics and enhancing feature expressiveness. The SSM outputs are fused with CNN-enhanced features to produce $\mathbf{F}_{xm1}$ and $\mathbf{F}_{xm2}$, which jointly capture temporal dependencies and local receptive field information. Finally, these features are integrated via the Mamba-Fusion module to produce the fused representation $\mathbf{F}_{\text{fusion}}$.

Text, as a high-level semantic carrier, enhances the discriminative power of multimodal features by introducing category descriptions to supplement semantic information, especially in scenarios with missing modality. To ensure the stability of semantic guidance, this article adopts a static prompt design, where all text prompts are generated offline before training and remain unchanged throughout the training process, with each category corresponding to a unique description. This design leverages the relatively stable semantic characteristics of remote sensing categories, avoiding the uncertainty brought about by dynamic prompt learning and reducing model complexity and computational overhead. In constructing the text prompts, structured semantic templates such as "category, color, shape, material, and height" are expanded into natural language descriptions using the generative pretrained transformer (GPT) series of large language models, and their semantic rationality and scene adaptability are ensured through manual screening and verification. This offline process does not participate in model training and inference, thus avoiding the introduction of additional computational costs.

In order to achieve the alignment of visual features and semantic features and thus guide feature transfer, a cross-modal contrastive loss function based on CLIP is introduced. Specifically, each category's description is encoded into textual features $\mathbf{F}_{\text{text}}$. A bidirectional similarity matrix is computed between the fused image features and the corresponding text features to measure their semantic correspondence, enabling the model to distinguish matched pairs from mismatched ones. This similarity serves as the foundation for the subsequent contrastive loss, which strengthens cross-modal feature alignment, formulated as

$$\mathbf{S}_{\text{fusion,text}} = \lambda_s \mathbf{F}_{\text{fusion}} \cdot \mathbf{F}_{\text{text}}^{\top} \tag{8}$$

$$\mathbf{S}_{\text{text,fusion}} = \mathbf{S}_{\text{fusion,text}}^{\top} \tag{9}$$

$$\begin{aligned} \mathcal{L}_{\text{clip}} = \frac{1}{2n} \sum_{i=1}^{n} \Big[&\text{CE}\left(\mathbf{S}_{\text{fusion,text}}^{(i)}, y^{(i)}\right) \\ &+ \text{CE}\left(\mathbf{S}_{\text{text,fusion}}^{(i)}, y^{(i)}\right) \Big] \end{aligned} \tag{10}$$

where $\mathbf{S}$ is the bidirectional similarity matrix, λ_s is an adjustable scaling factor, n is the number of categories and CE$(\cdot)$ is the cross-entropy loss. By optimizing this contrastive loss, the model achieves stronger cross-modal consistency, alleviates modality heterogeneity, and facilitates more effective knowledge transfer.

D. Training Objective

1) Feature Completion: In the SCFC, in order to effectively supplement the missing modality features, a joint objective function is proposed, which combines shared feature consistency and diffusion completion loss. This enhances both semantic consistency and feature discriminability. The total loss $\mathcal{L}_1$ is defined as

$$\mathcal{L}_1 = \mathcal{L}_{\text{diff}} + \lambda_{\text{sfc}} \mathcal{L}_{\text{sfc}} \tag{11}$$

where $\mathcal{L}_{\text{diff}}$ is the diffusion completion loss, $\mathcal{L}_{\text{sfc}}$ is the shared feature consistency loss, and λ_{sfc} is the weight of the shared feature consistency loss.

A diffusion-based reconstruction loss is employed to measure the discrepancy between the predicted and ground-truth latent features during the reverse diffusion process, where the model aims to reconstruct the original feature $\hat{\mathbf{Z}}_0$ from the noisy representation $\mathbf{Z}_T$. The loss $\mathcal{L}_{\text{diff}}$ is formulated as

$$\mathcal{L}_{\text{diff}} = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{\mathbf{Z}}_{0,i} - \mathbf{Z}_{0,i} \right\|_1 \tag{12}$$

where $\hat{\mathbf{Z}}_{0,i}$ denotes the reconstructed feature obtained from the reverse diffusion process, $\mathbf{Z}_{0,i}$ represents the corresponding ground-truth latent feature, and N denotes the total number of elements involved in the computation.

The shared feature consistency loss is introduced to constrain the high-order statistics of different modal features, enhance their expression consistency in the shared semantic space, and achieve more robust cross-modal alignment. The loss function is as follows:

$$\mathcal{L}_{\text{sfc}} = \sum_{k=1}^{K} \left\| c_k(\mathbf{F}_{x_{m1}}) - c_k(\mathbf{F}_{x_{m2}}) \right\|_2 \tag{13}$$

where $\mathbf{F}_{x_{m1}}$ and $\mathbf{F}_{x_{m2}}$ are the features of two modalities, c_k is the kth central moment, and K is the calculated central moment order.

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2) Dual Distillation: In the TGD^2, in order to effectively guide the student network to perform feature learning and classification optimization, a joint training objective is proposed that integrates three optimization signals: distillation of intermediate layer features, classification supervision of the student network's own predictions through cross entropy loss, and classification label distillation of the output layer. The overall loss $\mathcal{L}_2$ is expressed as

$$\mathcal{L}_2 = \alpha\mathcal{L}_{\text{logit}} + (1-\alpha)\mathcal{L}_{\text{CE}} + \beta\mathcal{L}_{\text{feat}} \tag{14}$$

where $\mathcal{L}_{\text{logit}}$ is the output-level distillation loss, $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss for supervising student predictions, $\mathcal{L}_{\text{feat}}$ is the feature-level distillation loss, and α and β are weight coefficients.

At the intermediate layer, the full multimodal features are used as guidance, and their feature representations are aligned using the mean squared error loss to approximate richer and more accurate semantic information. The feature-level distillation loss is defined as follows:

$$\mathcal{L}_{\text{feat}} = \|\mathbf{F}_{\text{s}} - \mathbf{F}_{\text{t}}\|_2^2 \tag{15}$$

where $\mathbf{F}_{\text{s}}$ represents the fused features obtained by integrating the incomplete modality features with the reconstructed missing-modal features, and $\mathbf{F}_{\text{t}}$ denotes the complete multimodal features obtained from image–text alignment in the teacher network.

At the output level, knowledge distillation is performed by aligning the prediction distributions of the complete and incomplete networks, smoothing the outputs using a temperature coefficient T_c, and minimizing their Kullback–Leibler divergence to guide the behavior of mimicking the complete multimodal model. The output distillation loss is defined as

$$\mathcal{L}_{\text{logit}} = \text{KL}\left(\log \text{Softmax}\left(\frac{\mathbf{s}}{T_c}\right), \text{Softmax}\left(\frac{\mathbf{t}}{T_c}\right)\right) {T_c}^2. \tag{16}$$

where $\mathbf{s}$ denotes the predictions from fused incomplete and reconstructed features, and $\mathbf{t}$ denotes those from complete multimodal features.

Simply relying on distillation signals may not be sufficient to ensure the classification ability of the student network. Therefore, the cross entropy loss $\mathcal{L}_{\text{CE}}$ is introduced to enhance the discriminative feature learning through explicit supervision of the real labels, thereby improving the classification performance.

IV. Experiments

A. Datasets Description

In this article, three publicly available multimodal remote sensing datasets are used to evaluate the effectiveness of the proposed classification method, particularly under scenarios with missing modality. These datasets span different sensing modalities and land cover types, providing a comprehensive benchmark for assessing the robustness and generalization capability of the proposed approach. Detailed descriptions of each dataset are given below.

TABLE I
Detailed Categories and Sample Size of the Houston2013 Dataset

Category		Number of samples	
No	Name	Train	Test
1	Healthy grass	40	1251
2	Stressed grass	40	1254
3	Synthetic grass	40	697
4	Trees	40	1244
5	Soil	40	1242
6	Water	40	325
7	Residential	40	1268
8	Commercial	40	1244
9	Road	40	1252
10	Highway	40	1227
11	Railway	40	1235
12	Parking Lot 1	40	1233
13	Parking Lot 2	40	469
14	Tennis court	40	428
15	Running track	40	660
Total		600	15029

TABLE II
Detailed Categories and Sample Size of the Trento Dataset

Category		Number of samples	
No	Name	Train	Test
1	Trees	40	4034
2	Buildings	40	2903
3	Ground	40	479
4	Woods	40	9123
5	Vineyard	40	10501
6	Roads	40	3174
Total		240	30214

1) Houston2013: The Houston2013 dataset was acquired by an ITRES CASI-1500 sensor over the University of Houston campus and surrounding areas for the 2013 IEEE International Geoscience and Remote Sensing Symposium (IGARSS) data fusion competition. The dataset comprises hyperspectral (HS), multispectral (MS), and LiDAR images, including 15 029 labeled samples distributed across 15 land cover classes, which makes it suitable for multimodal remote sensing classification tasks, as detailed in Table I.

2) Trento: The Trento dataset is collected over a rural area in Trento, Southern Italy, and includes HS and LiDAR data. The HS data, acquired by an AISA Eagle sensor, spans 402.89–989.09 nm with 63 bands, while the LiDAR data is collected using an Optech ALTM 3100EA sensor. It contains 30 214 labeled samples, covering six land cover classes, as detailed in Table II.

3) Augsburg: The Augsburg dataset originates from the Augsburg region of Germany and consists of HS data, LiDAR data, and synthetic aperture radar (SAR) data. The HS data contains 180 spectral bands, the SAR data has four channels, and the LiDAR data is a single-channel gray-scale image. The dataset images have a size of 332×485 pixels and contain 78 294 labeled samples covering seven different land-cover categories, as detailed in Table III.

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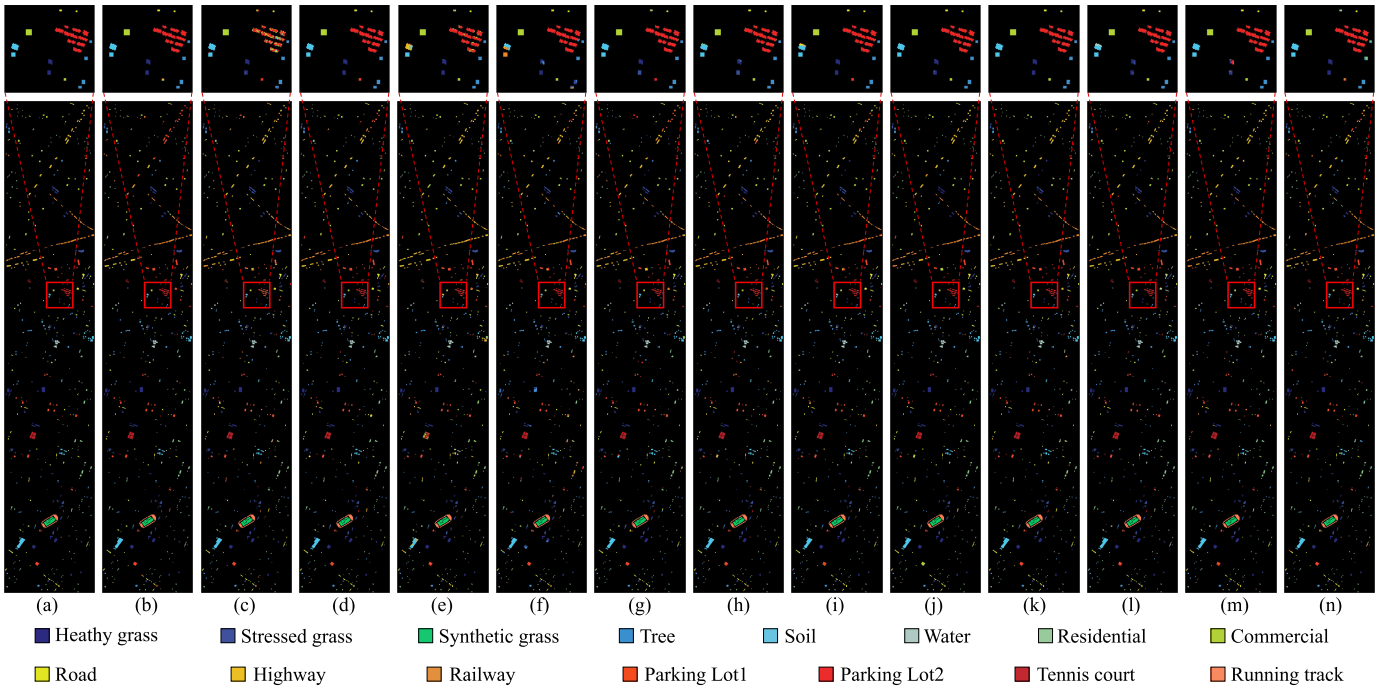


Fig. 4. Classification maps of the Houston dataset. (a) Ground truth. (b) HS-Net(Ours). (c) MDL-RS. (d) MSH-Net. (e) DGD-Net. (f) MCMoE. (g) GMKFN. (h) DFC-TD^2 (HS). (i) M2FNet. (j) MSFE-IFN. (k) Sal^{2}RN. (l) S3F2Net. (m) MSFMamba. (n) DFC-TD^2 (HS and LiDAR).

TABLE III
DETAILED CATEGORIES AND SAMPLE SIZE OF THE AUGSBURG DATASET

Category		Number of samples	
No	Name	Train	Test
1	Forest	40	13507
2	Residential Area	40	30329
3	Industrial Area	21	3851
4	Low Plants	40	26857
5	Allotment	40	575
6	Commercial Area	7	1645
7	Water	23	1530
Total		211	78294

B. Experiments Setup

1) Evaluation Index: Three classification evaluation indices are considered, which are overall accuracy (OA), average accuracy (AA), and kappa coefficient (Kappa).

2) Implementation Details: The proposed method is implemented on the PyTorch platform and trained on one NVIDIA GeForce 2080 GPU. The entire framework is trained using the Adam optimizer, with 500 training epochs, a modality compensation time step of 1000, a batch size of 1024, and a learning rate of $1e^{-4}$.

3) Comparison Methods: To verify the effectiveness of the proposed multimodal remote sensing image joint classification method in the presence of missing modality, this article designs three different experimental configurations as follows.

1) In the proposed method, training and testing are both performed within a single modality, including MDL-RS [43]. For example, "HS-net" represents a model trained and tested using only HS images; "LiDAR/SAR" represents a model trained and tested using only LiDAR or SAR images.

2) State-of-the-art methods for modality-missing scenarios, including MSH-Net [44], DGD-Net [40], MCMoE [45], and GMKDN [46]. "W/o" indicates the case where the corresponding modality is missing in inference.

3) State-of-the-art methods for multimodal remote sensing joint classification, including M2FNet [47], MSFE-IFN [48], Sal^{2}RN [49], S3F2Net [50], and MSFMamba [32]. To ensure the fairness and reproducibility of the experimental results, all comparison methods are retrained on the same GPU computing environment and deep learning framework version. The parameter settings for each method remained consistent with the original paper, and they were trained until convergence to obtain their optimal test performance. All experimental results are presented in the classification performance statistics table and classification result visualization.

C. Comparison With State-of-the-Art Methods

1) Results and Analysis of HS-LiDAR: To verify the effectiveness of the proposed DFC-TD^2 method, this article conducted systematic comparative experiments on multiple datasets, and the relevant quantitative results are summarized in Table IV. Meanwhile, to visually demonstrate the classification performance of different methods, classification visualization results on the Houston and Trento datasets are presented, as shown in Figs. 4 and 5, respectively. Under the single-modal training and testing setting, the proposed method outperforms the compared methods in terms of OA, AA, and the Kappa coefficient. This indicates that DFC-TD^2 can more effectively mine discriminative features, possessing stronger feature modeling and classification capabilities, thus exhibiting significant performance advantages in single-modal scenarios.

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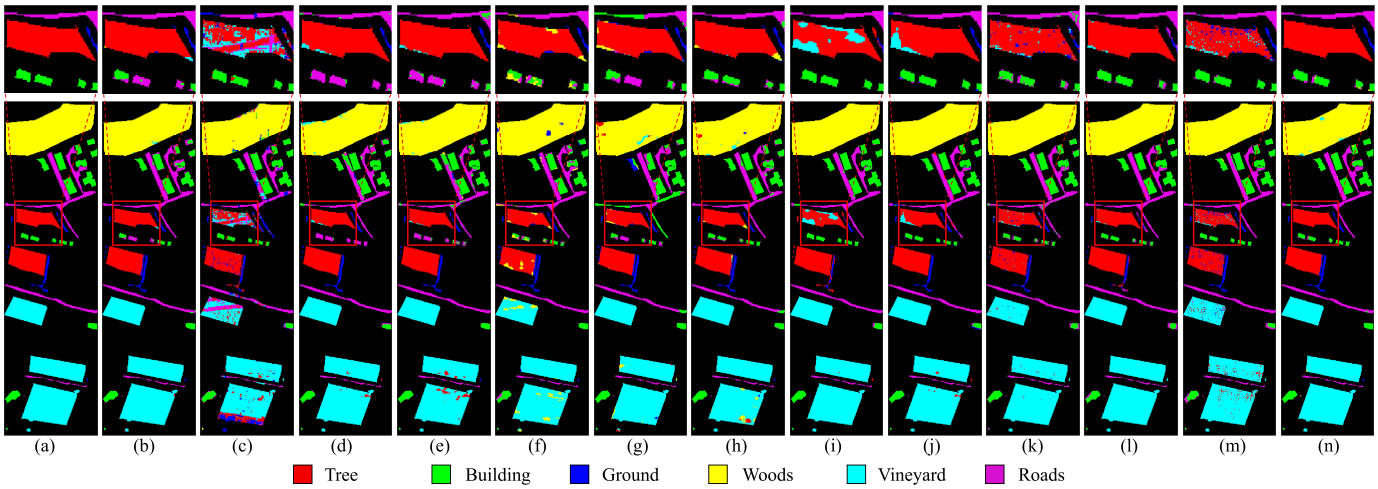


Fig. 5. Classification maps of the Trento dataset. (a) Ground truth. (b) HS-Net(Ours). (c) MDL-RS. (d) MSH-Net. (e) DGD-Net. (f) MCMoE. (g) GMKFN. (h) DFC-TD^2 (HS). (i) M2FNet. (j) MSFE-IFN. (k) Sal^{2}RN. (l) S3F2Net. (m) MSFMamba. (n) DFC-TD^2 (HS and LiDAR).

TABLE IV

CLASSIFICATION ACCURACY OF VARIOUS METHODS ON THE HOUSTON2013, TRENTO, AND AUGSBURG DATASETS. "HS-NET(OURS)" REFERS TO EXPERIMENTS CONDUCTED USING ONLY HS IMAGES. "H, L" AND "H, S" INDICATE EXPERIMENTS USING HS WITH LIDAR AND SAR IMAGES, RESPECTIVELY. "W/O" INDICATES THE MISSING MODALITY IN INFERENCE

Method	Houston2013					Trento					Augsburg				
	Train	Test	OA (%)	AA (%)	Kappa	Train	Test	OA (%)	AA (%)	Kappa	Train	Test	OA (%)	AA (%)	Kappa
	single modality					**single modality**					**single modality**				
MDL-RS	HS	HS	84.42	85.30	83.16	HS	HS	86.06	83.08	81.79	HS	HS	62.34	64.65	52.16
HS-Net(Ours)	HS	HS	94.05	95.12	93.56	HS	HS	93.17	93.20	90.86	HS	HS	85.26	83.36	79.96
MDL-RS	LiDAR	LiDAR	50.74	51.22	47.15	LiDAR	LiDAR	76.85	79.48	70.06	SAR	SAR	55.70	55.20	45.37
LiDAR/SAR-Net(Ours)	LiDAR	LiDAR	56.91	61.70	53.85	LiDAR	LiDAR	92.98	92.23	90.76	SAR	SAR	75.80	72.94	74.77
	W/o HS modality					**W/o HS modality**					**W/o HS modality**				
MSH-Net	H , L	LiDAR	58.38	60.90	55.57	H , L	LiDAR	89.81	86.66	87.90	H , S	SAR	62.95	55.32	58.23
DGD-Net	H , L	LiDAR	55.92	57.71	52.98	H , L	LiDAR	83.78	82.60	80.24	H , S	SAR	59.41	58.76	51.16
MCMoE	H , L	LiDAR	66.31	63.78	63.62	H , L	LiDAR	88.61	83.26	84.87	H , S	SAR	79.63	54.97	71.18
GMKDN	H , L	LiDAR	61.93	66.09	59.12	H , L	LiDAR	89.39	86.58	86.02	H , S	SAR	72.47	62.12	63.80
DFC-TD^2(Ours)	H , L	LiDAR	**85.71**	**88.03**	**84.58**	H , L	LiDAR	**96.87**	**96.54**	**95.83**	H , S	SAR	**84.65**	**78.30**	**77.82**
	W/o LiDAR modality					**W/o LiDAR modality**					**W/o SAR modality**				
MSH-Net	H , L	HS	92.31	93.62	91.80	H , L	HS	95.62	93.45	94.79	H , S	HS	81.24	69.90	79.00
DGD-Net	H , L	HS	83.15	85.90	81.95	H , L	HS	95.77	94.52	94.56	H , S	HS	82.63	74.08	80.37
MCMoE	H , L	HS	83.93	86.32	82.64	H , L	HS	94.50	94.05	92.67	H , S	HS	81.96	78.81	75.49
GMKDN	H , L	HS	87.40	89.74	86.40	H , L	HS	93.22	90.23	90.98	H , S	HS	82.23	71.89	75.32
DFC-TD^2(Ours)	H , L	HS	**94.82**	**95.01**	**93.89**	H , L	HS	**96.20**	**95.69**	**94.92**	H , S	HS	**87.15**	**80.96**	**80.42**
	complete modalities					**complete modalities**					**complete modalities**				
M2FNet	H , L	H , L	93.57	94.55	93.05	H , L	H , L	97.12	97.94	96.14	H , S	H , S	88.49	78.92	83.87
MSFE-IFN	H , L	H , L	94.01	94.90	93.58	H , L	H , L	98.29	97.43	97.73	H , S	H , S	89.69	81.70	86.91
Sal^{2}RN	H , L	H , L	94.08	95.27	94.52	H , L	H , L	97.59	94.83	96.78	H , S	H , S	82.53	79.95	87.35
S3F2Net	H , L	H , L	93.52	94.63	93.01	H , L	H , L	98.54	92.23	98.05	H , S	H , S	89.62	82.86	85.56
MSFMamba	H , L	H , L	94.62	95.49	94.19	H , L	H , L	95.59	95.10	94.16	H , S	H , S	87.73	80.64	83.06
DFC-TD^2(Ours)	H , L	H , L	**95.11**	**96.02**	**94.71**	H , L	H , L	**98.72**	**98.10**	**98.29**	H , S	H , S	**90.45**	**85.85**	**87.36**

In the case of missing modality, the proposed DFC-TD^2 method exhibits significant performance advantages. Taking the Houston dataset as an example, when the HS is missing, the OA of DFC-TD^2 is improved by 27.33%, 4.47%, 19.10%, and 23.78% compared to MSH-Net, DGD-Net, MCMoE, and GMKDN, respectively. When the LiDAR is missing, the OA

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TABLE V
ACCURACY FOR EACH CLASS (%) ON THE HOUSTON2013 DATASET UNDER MISSING LIDAR MODALITY

Category	MSH-Net	DGD-Net	MCMoE	GMDKN	DKD-TD^2
Healthy grass	93.37	88.65	61.79	94.88	97.76
Stressed grass	99.84	99.92	86.52	95.06	98.41
Residential	91.56	63.33	68.85	85.57	94.95
Commercial	81.19	100.00	77.33	72.35	71.46
Road	85.06	64.30	78.91	79.79	80.91
Highway	82.07	89.57	89.41	72.13	94.70
Parking Lot1	88.97	99.68	84.75	72.67	83.54
Parking Lot2	95.74	73.35	92.32	99.57	93.60
Tennis court	100.00	58.41	99.07	100.00	100.00
Running track	100.00	100.00	100.00	98.79	100.00

of DFC-TD^2 is still improved by 2.51%, 11.67%, 10.89%, and 7.42% compared to the aforementioned methods, respectively. Furthermore, experimental results on the Trento dataset also validate the robustness and effectiveness of DFC-TD^2 under missing modality. In addition, as shown in Table V, five pairs of categories with similar spectral features are selected from the Houston2013 dataset, and OA (%) are compared under the missing LiDAR modality setting, where the proposed DKD-TD^2 achieves clear advantages in several easily confused categories. Specifically, for vegetation categories (Healthy grass and Stressed grass), the accuracies reach 97.76% and 98.41%, respectively, demonstrating their ability to capture subtle spectral differences, while for transportation infrastructure, the accuracy of Highway reaches 94.70%, effectively reducing confusion with road. In addition, the method achieves 100.00% accuracy on both the tennis court and the running track, highlighting its strong discriminative capability. The comprehensive quantitative results show that DFC-TD^2 can not only effectively supplement and reconstruct key discriminative features in the case of missing modality, but also achieve full transfer and constraint of cross-modal knowledge by introducing a text-guided DD mechanism, thus maintaining stable and superior classification performance when relying only on partial modal inputs.

2) Results and Analysis of HS-SAR: This article conducts systematic experiments on the Augsburg dataset based on two modes: HS and SAR. Table IV presents the comparison results of different methods on evaluation metrics such as OA, AA, and the Kappa coefficient. The corresponding classification visualization results are shown in Fig. 6.

In the case of missing modality, DFC-TD^2 still exhibits significant performance advantages. When HS is missing, the OA of DFC-TD^2 is improved by 21.70%, 25.24%, 5.02%, and 12.18% compared to MSH-Net, DGD-Net, MCMoE, and GMKDN, respectively. When the SAR mode is missing, its OA is still improved by 5.91%, 4.52%, 5.19%, and 4.92% compared to the above methods, respectively. These results fully demonstrate that the proposed method still possesses good robustness and stable classification performance even when key modal inputs are missing. Furthermore, in the case of full-modal setting, DFC-TD^2 also achieved competitive results, with its OA improving by 1.96%, 0.76%, 7.92%,

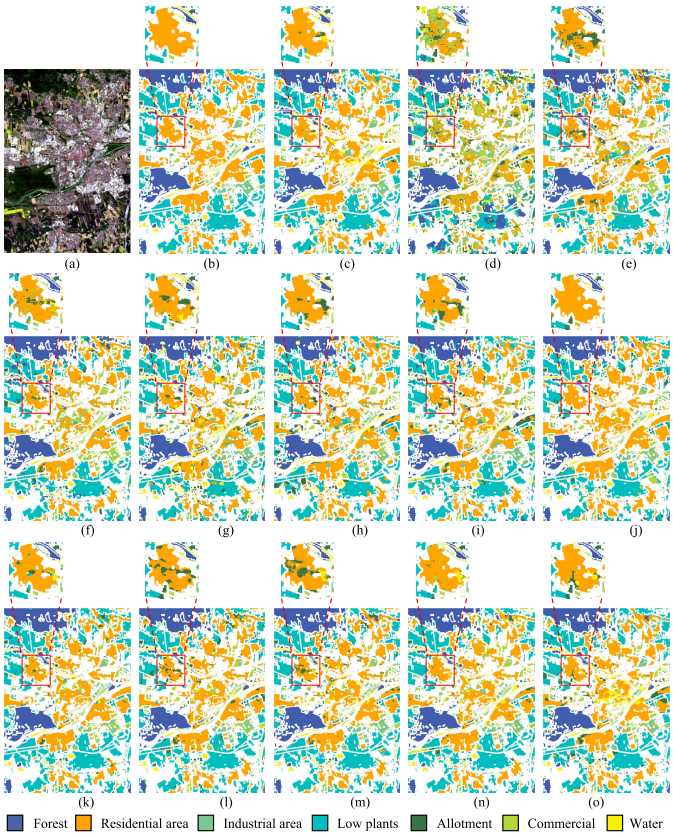


Fig. 6. Classification maps of the Augsburg dataset. (a) Augsburg(HS). (b) Ground truth. (c) HS-Net(Ours). (d) MDL-RS. (e) MSH-Net. (f) DGD-Net. (g) MCMoE. (h) GMKDN. (i) DFC-TD^2(HS). (j) M2FNet. (k) MSFE-IFN. (l) Sal^{2}RN. (m) S3F2Net. (n) MSFMamba. (o) DFC-TD^2(HS, SAR).

and 2.72% compared to M2FNet, MSFE-IFN, Sal^{2}RN, and MSFMamba, respectively. These results further validate the effectiveness of DFC-TD^2 in multimodal feature modeling and information fusion, fully leveraging the complementary information between different modalities to achieve more accurate and robust remote sensing image classification.

3) Results and Analysis of HS-SAR-LiDAR: This section constructs two experimental settings on the Augsburg dataset for HS, SAR, and LiDAR modalities, one for single-modal missing data and the other for multimodal missing data, to systematically evaluate the classification performance of the proposed DFC-TD^2 method. Table VI summarizes the quantitative comparison results of different methods under the above experimental settings. The experimental results show that compared with the state-of-the-art model, DFC-TD^2 achieves an OA improvement of 0.75%–18.72% under the single-modal missing condition, and still maintains a classification accuracy of over 81.44% under the condition of simultaneous multimodal missing data, fully demonstrating its robustness and good generalization ability in complex multimodal missing scenarios.

Further analysis reveals that the superior performance of DFC-TD^2 mainly stems from the synergistic effect of two aspects. On the one hand, the diffusion FC mechanism effectively alleviates the problem of incomplete information caused by modality loss. On the other hand, the text-guided DD strategy promotes cross-modal alignment and knowledge

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TABLE VI

JOINT CLASSIFICATION ACCURACY OF COMBINATIONS OF HS, SAR, AND LIDAR ON THE AUGSBURG DATASET

Method	Train	Test	OA(%)	AA(%)	Kappa
MSH-Net	HS,SAR,LiDAR	HS	86.93	79.37	81.45
DFC-TD^2	HS,SAR,LiDAR	HS	**87.68**	**84.87**	**82.14**
MSH-Net	HS,SAR,LiDAR	SAR	61.75	55.18	56.60
DFC-TD^2	HS,SAR,LiDAR	SAR	**77.02**	**76.43**	**79.72**
MSH-Net	HS,SAR,LiDAR	LiDAR	55.19	55.31	48.65
DFC-TD^2	HS,SAR,LiDAR	LiDAR	**73.91**	**74.60**	**75.82**
MSH-Net	HS,SAR,LiDAR	HS, SAR	87.48	78.73	82.74
DFC-TD^2	HS,SAR,LiDAR	HS, SAR	**89.83**	**86.02**	**85.87**
MSH-Net	HS,SAR,LiDAR	HS, LiDAR	89.11	79.49	84.87
DFC-TD^2	HS,SAR,LiDAR	HS, LiDAR	**89.28**	**85.50**	**85.16**
MSH-Net	HS,SAR,LiDAR	LiDAR, SAR	63.40	57.83	53.59
DFC-TD^2	HS,SAR,LiDAR	LiDAR,SAR	**81.44**	**81.11**	**75.19**

TABLE VII

COMPARISON OF COMPUTATIONAL COMPLEXITY AND INFERENCE EFFICIENCY

Method	Train		Test		
	Params (M)	FLOPs (G)	Time (ms)	Params (M)	FLOPs (G)
DGD-Net	0.60	0.024	1.32	0.60	0.024
GMKDN	0.40	0.048	2.94	5.16	0.057
Ours	0.78	0.054	5.24	30.95	0.863

transfer, maintaining stable performance even when multiple modalities are lost simultaneously.

4) Computational Complexity and Efficiency Analysis: We quantitatively compared the proposed method with state-of-the-art approaches, including DGD-Net and GMKDN, in terms of parameter size, FLOPs, and inference time, as shown in Table VII. Owing to the integration of the latent diffusion-based FC module, the Mamba-based bidirectional encoder and fusion module, and the text-guided distillation mechanism, the proposed method introduces additional computational overhead. Nevertheless, these components are crucial for explicitly modeling missing modalities, exploiting complementary information across modalities, and facilitating cross-modal feature transfer, thereby enhancing the richness and discriminative power of feature representations. Despite the increased complexity, the model achieves an inference time of only 5.24 ms, meeting the requirements of remote sensing applications and demonstrating its practical feasibility. More importantly, the additional cost yields consistent improvements in classification accuracy and robustness, resulting in a favorable tradeoff between performance and efficiency compared with existing methods.

D. Ablation Experiments

1) Effectiveness of Textual Guidance (TG): To evaluate the effective guiding role of text embedding in knowledge transfer during feature distillation, this article conducts an ablation experiment. While keeping other network structures and training strategies unchanged, text guidance information is removed from the intermediate feature layers. The experimental results are summarized in Fig. 7. Taking the Houston dataset as an example, with the HS modality missing, the introduction of text guidance improved the OA, AA, and Kappa coefficient by 5.11%, 4.26%, and 5.51%, respectively. With the LiDAR modality missing, text guidance also brought significant performance improvements, increasing OA, AA, and Kappa coefficients by 0.57%, 0.2%, and 0.11%, respectively.

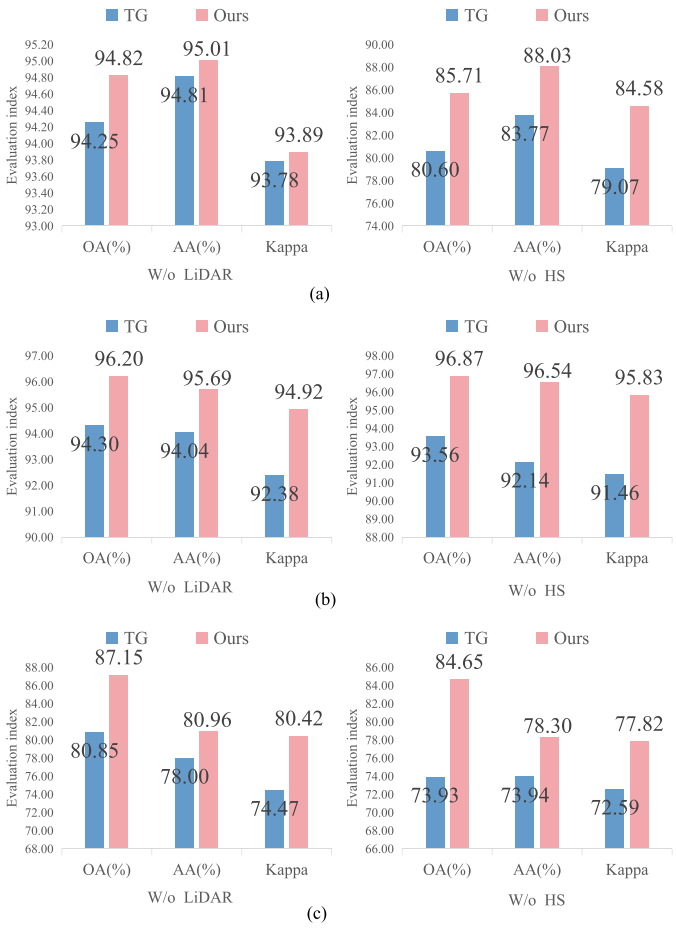


Fig. 7. Classification results for evaluating the effectiveness of text-guided feature knowledge transfer on three datasets, where "W/o" denotes the missing modality during inference. (a) Houston. (b) Trento. (c) Augsburg.

These results demonstrate that incorporating text embeddings into the feature distillation process provides high-level semantic guidance, enabling the model to focus on more discriminative regions and representations, thereby alleviating information loss caused by missing modalities. By leveraging text-guided distillation, semantic knowledge can be more effectively transferred, mitigating the adverse effects of modal heterogeneity and substantially improving feature representation and classification performance.

2) Effectiveness of DD: To verify the effectiveness of intermediate layer feature distillation in the proposed framework, ablation experiments were conducted. While maintaining other network structures and training strategies, the intermediate layer feature distillation module is removed, retaining only the output layer probabilistic distillation mechanism. Experimental results are shown in Fig. 8. On the Trento dataset, when the HS modality is missing, intermediate layer feature distillation

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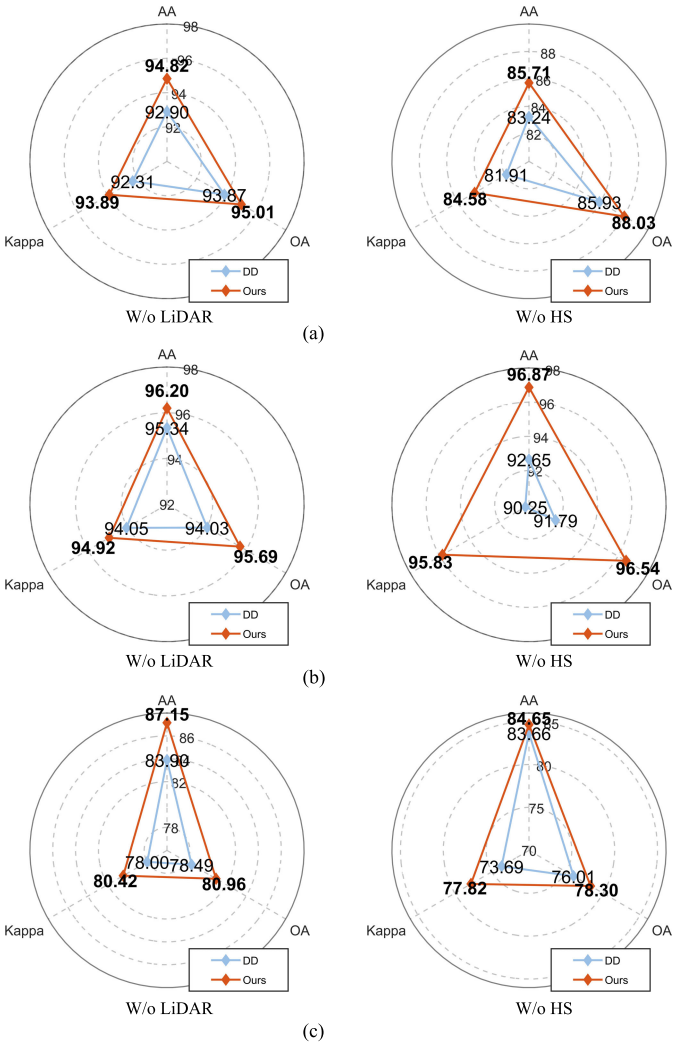


Fig. 8. Classification results for evaluating the effectiveness of DD on three datasets, where "W/o" denotes the missing modality during inference. (a) Houston. (b) Trento. (c) Augsburg.

significantly improves model performance, increasing OA, AA, and Kappa coefficients by 4.22%, 4.75%, and 5.58%, respectively. When the LiDAR modality is missing, compared to the scheme using only output layer probabilistic distillation, the DD strategy incorporating intermediate feature supervision still delivers stable performance gains, improving various metrics by 0.86%, 1.66%, and 0.87%, respectively.

These results show that intermediate feature distillation provides fine-grained, multilevel supervision, enabling the model to align not only with final predictions but also to effectively capture discriminative knowledge at the feature representation level. Compared with single-layer distillation based solely on output probability alignment, the proposed DD strategy exhibits superior robustness and stability under missing modality conditions, highlighting the critical role of intermediate feature supervision in alleviating performance degradation caused by missing modalities.

3) Effectiveness of Mamba-Based Encoder: To evaluate the effectiveness of the Mamba-based encoder in feature modeling, this article conducts ablation experiments in a full-modal setting, replacing it with representative CNN-based and transformer-based encoders for comparison. The experimental results are shown in Table VIII, demonstrating that the Mamba-based encoder consistently outperforms the CNN and Transformer encoders in classification accuracy. Although the ablation experiments are conducted in a full-modal configuration, this encoder primarily performs modality-specific feature modeling, and its mechanism is independent of subsequent modality completion or dual knowledge transfer strategies. Therefore, the superior representational capabilities exhibited by the Mamba-based encoder can be reasonably extended to scenarios with missing modality. Based on this observation, this article consistently uses the Mamba-based encoder as the backbone network for feature modeling in all experimental settings, regardless of whether the modality is complete.

TABLE VIII

CLASSIFICATION RESULTS FOR EVALUATING THE EFFECTIVENESS OF THE MAMBA-BASED ENCODER

Datasets	Method	OA(%)	AA(%)	Kappa
Houston	CNN	94.24	94.94	93.77
	Transformer	93.99	94.99	93.50
	Mamba	95.11	96.02	94.71
Trento	CNN	98.32	98.06	97.75
	Transformer	91.34	93.29	88.46
	Mamba	98.72	98.10	98.29
Augsburg	CNN	82.22	79.57	75.96
	Transformer	76.87	74.20	69.14
	Mamba	90.45	85.85	87.36

TABLE IX

CLASSIFICATION RESULTS FOR EVALUATING THE EFFECTIVENESS OF FC ON THREE DATASETS. "W/O" DENOTES THE MISSING MODALITY DURING INFERENCE

Datasets	Method	W/o LiDAR			W/o HS		
		OA(%)	AA(%)	Kappa	OA(%)	AA(%)	Kappa
Houston	FC	79.89	82.69	78.26	56.91	61.70	53.85
	Ours	94.82	95.01	93.89	85.71	88.03	84.58
Trento	FC	94.09	90.54	92.06	95.22	92.82	93.61
	Ours	96.20	95.69	94.92	96.87	96.54	95.83
Augsburg	FC	79.89	82.69	78.26	56.91	61.70	53.85
	Ours	94.82	95.01	93.89	85.71	88.03	84.58

4) Effectiveness of FC: To verify the effectiveness of the modality FC network within the overall framework, this article conducts ablation experiments. While maintaining the DD network structure, a shared-constrained diffusion FC module is removed, leaving only the distillation mechanism for training and testing. Experimental results are shown in Table IX. Taking the Augsburg dataset as an example, with the HS modality missing, the model shows significant gains in OA, AA, and Kappa metrics after FC, with particularly noticeable improvements in OA and Kappa. Similarly, with the SAR modality missing, all three metrics show stable improvement, indicating that the completion mechanism plays a positive role in different missing scenarios.

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Therefore, the proposed diffusion FC network can effectively recover discriminative information in the case of missing modality, thereby generating more complete and consistent feature representations for subsequent feature alignment and classification, thus improving the classification accuracy and robustness in complex multimodal remote sensing scenarios.

V. Conclusion

This work considers a commonly adopted setting where all modalities are available during training, enabling the learning of stable and semantically consistent cross-modal representations. Against this backdrop, we propose a DFC-TD^2 to tackle the common issue of missing modality in multimodal remote sensing image classification. The model first completes missing modality information via an SCFC, which takes shared features as conditional inputs and guides the potential diffusion to recover feature information. Then, a TGD^2 is introduced to align cross-modal features using textual semantics and to enhance knowledge transfer through joint supervision at both feature and output levels. Extensive comparative and ablation experiments are conducted on several representative remote sensing benchmark datasets, systematically validating the robustness and effectiveness of the proposed method in both modality-complete and modality-incomplete application scenarios. In more complex real-world scenarios, modality loss may also occur during the training phase, and extending the proposed framework to these settings would be an important direction for future research.

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